Old Chicago Main Post Office: Difference-in-Differences Analysis

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1 Post Office Building Value Trajectory

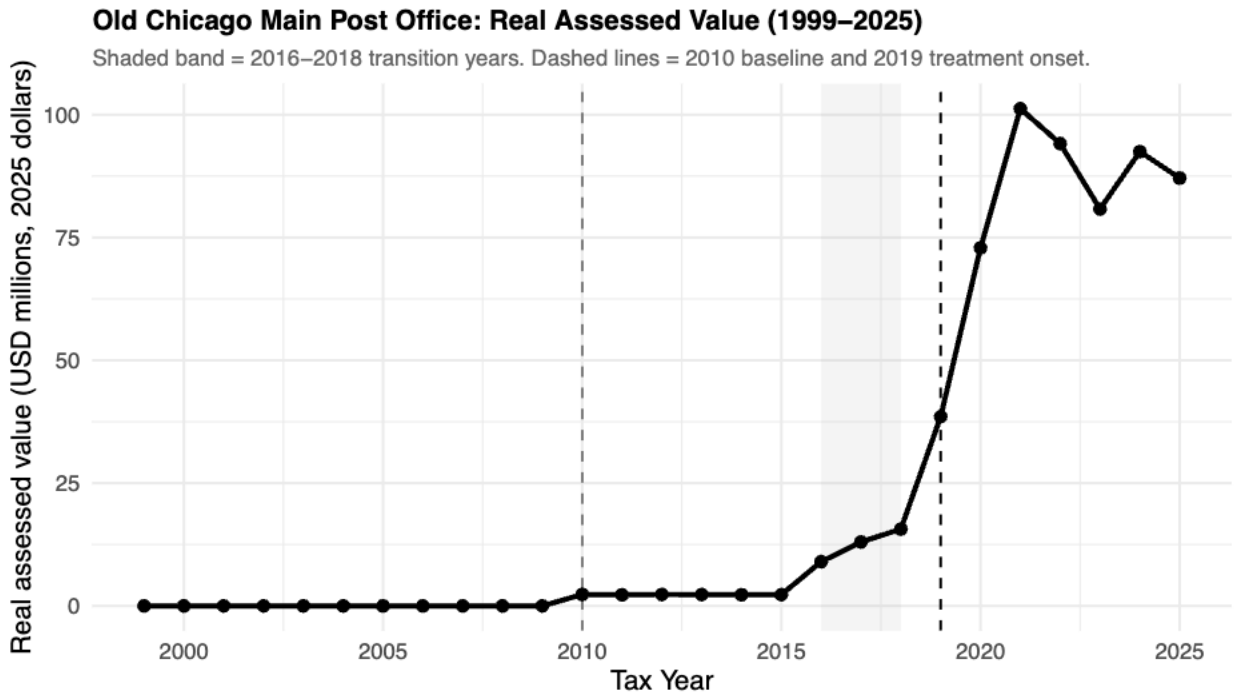

Figure 1: Real assessed value of the Old Chicago Main Post Office parcel, 1999-2025. Shaded band marks 2016-2018 transition period.

2 Parallel Trends

2.1 Formal pre-period: 1999-2015

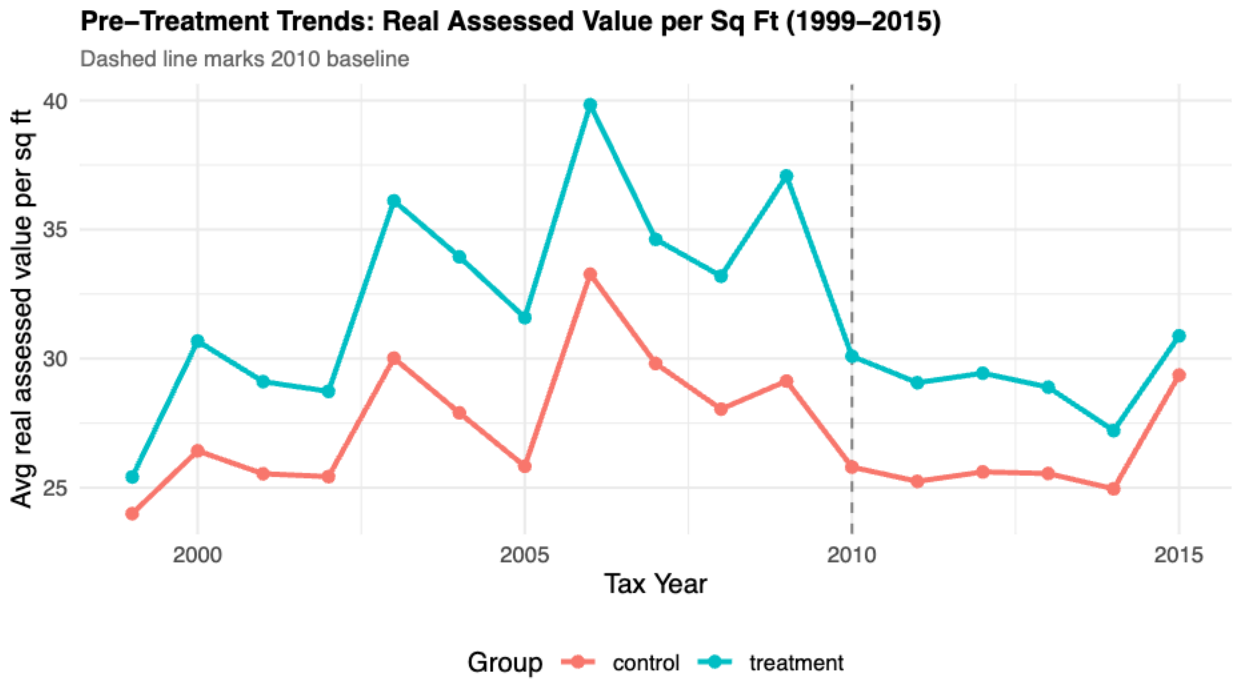

Figure 2: Average real assessed value per sq ft, treatment vs control, 1999-2015. Dashed line marks 2010 baseline.

2.2 Extended pre-period including transition years: 1999-2018

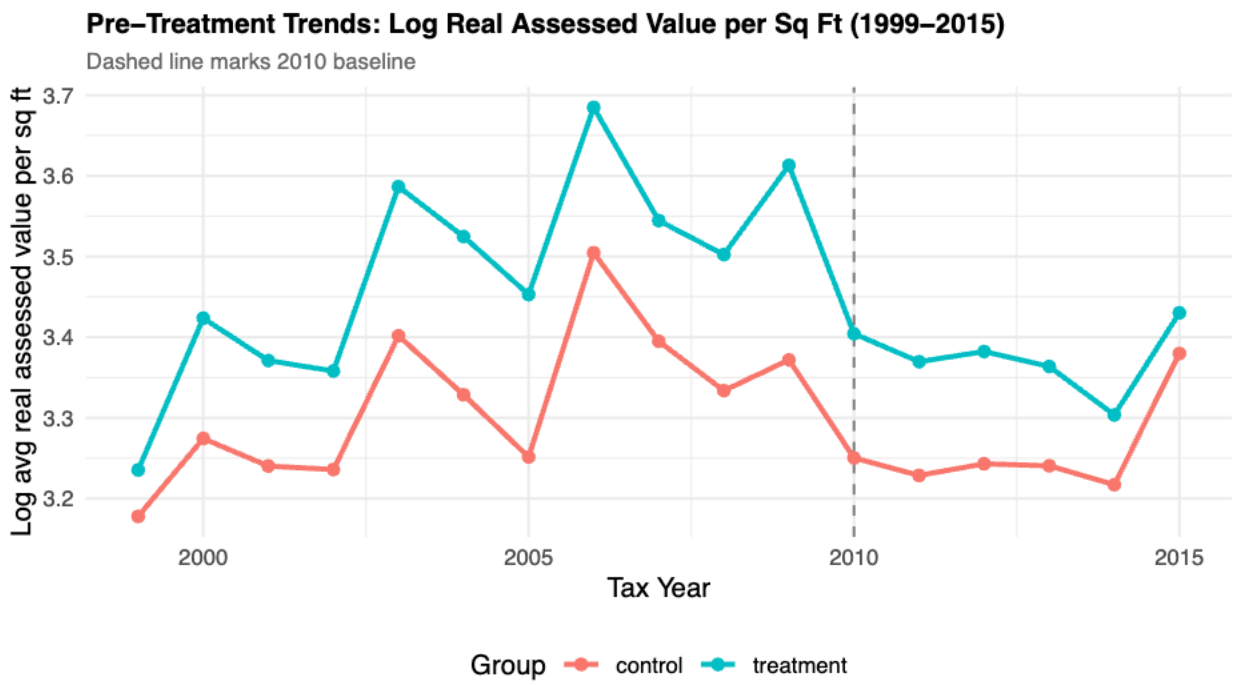

Figure 3: Log average real assessed value per sq ft, treatment vs control, 1999-2015.

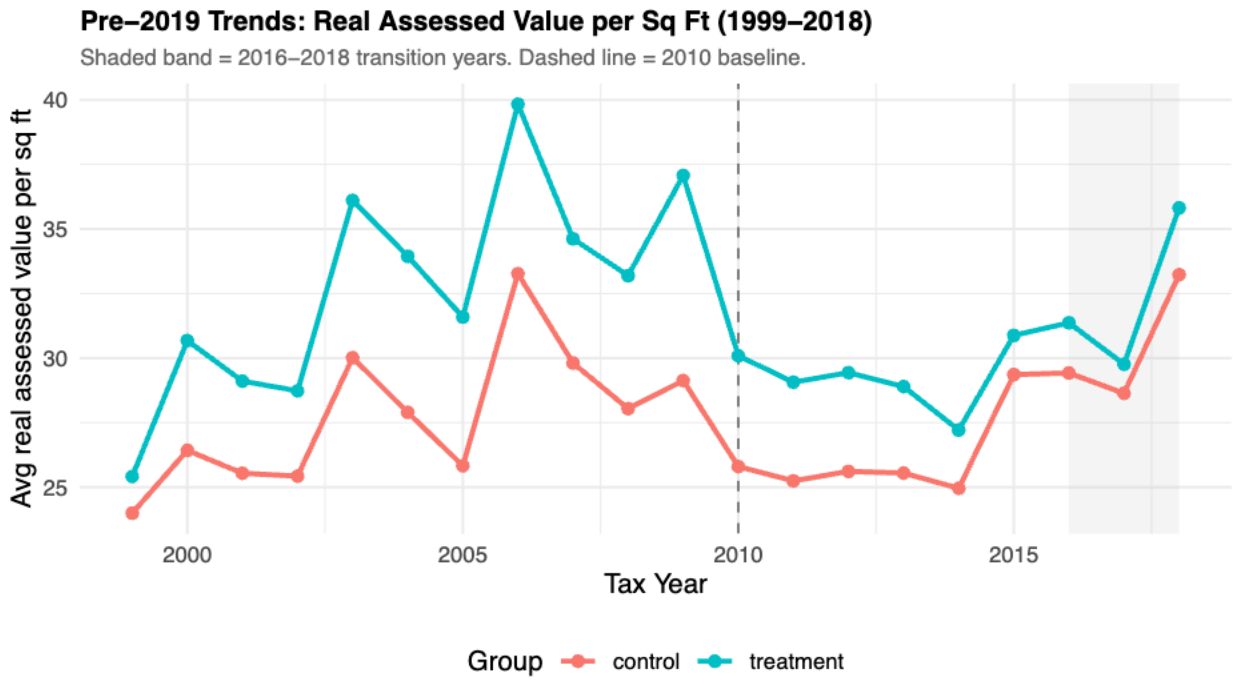

Figure 4: Average real assessed value per sq ft including 2016-2018 transition years. Shaded band marks transition period.

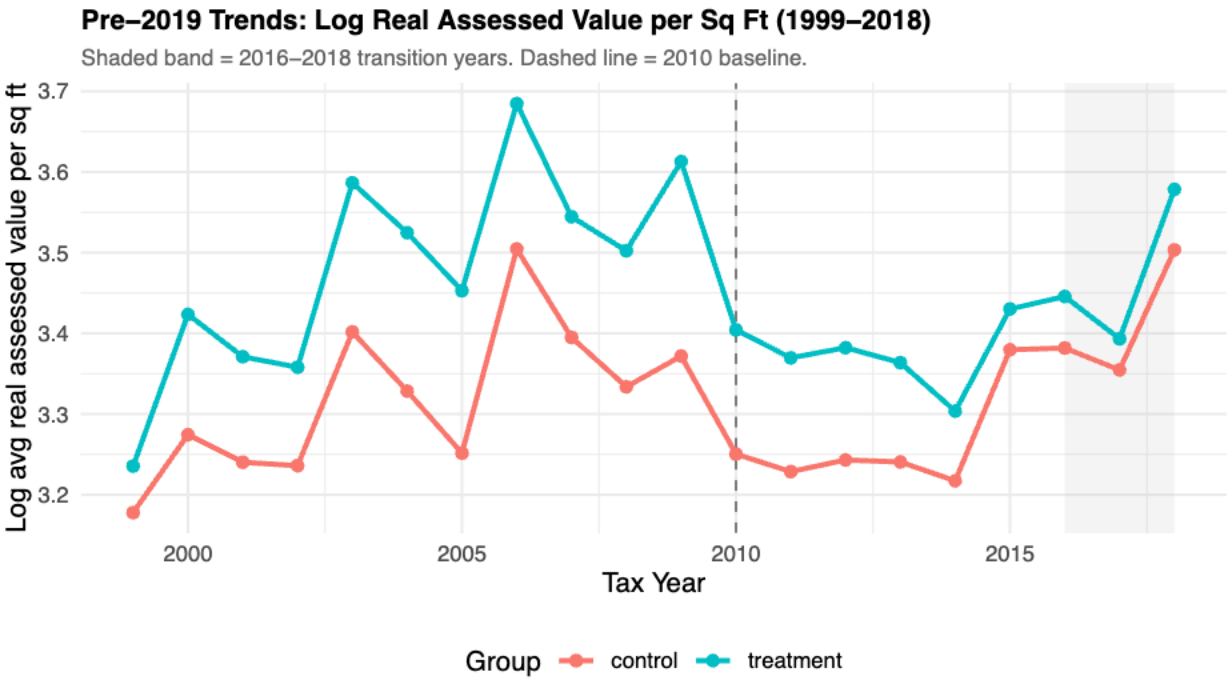

Figure 5: Log average real assessed value per sq ft including 2016-2018 transition years.

3 Full-Period Descriptive Trends (1999-2025)

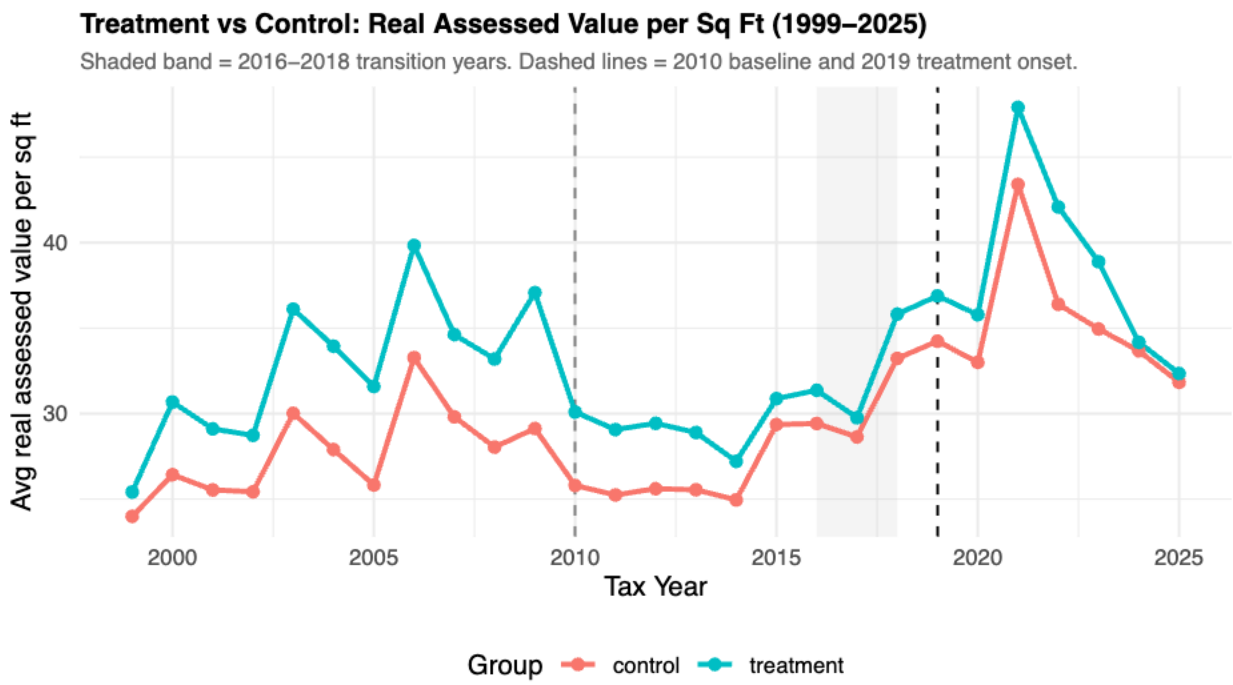

Figure 6: Full-period average real assessed value per sq ft. Shaded band = 2016-2018 transition years. Dashed lines = 2010 baseline and 2019 treatment onset.

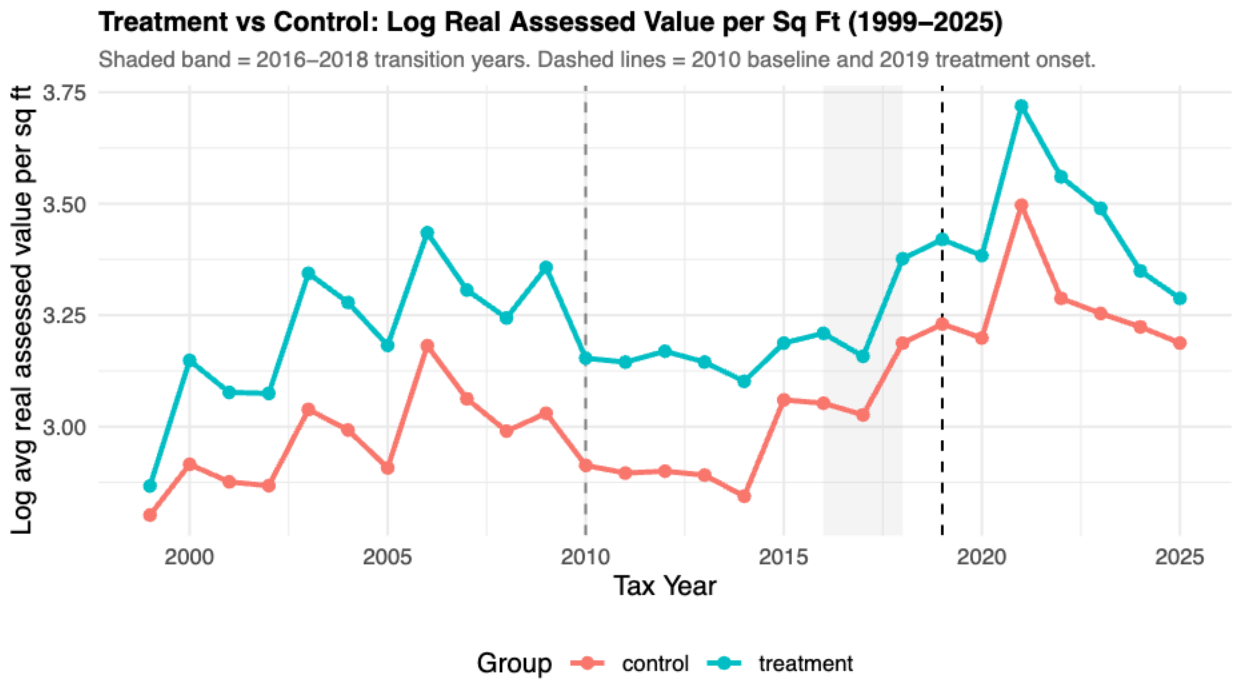

Figure 7: Full-period log average real assessed value per sq ft.

4 Difference-in-Differences Results

4.1 Two-Way Fixed Effects Models

```
## Main specification (all years):

## OLS estimation, Dep. Var.: log_value_psf
## Observations: 10,608
## Fixed-effects: pin: 458,  tax_year: 27
## Standard-errors: Clustered (pin)
##                 Estimate Std. Error   t value Pr(>|t|)
## treatment:post -0.025919   0.047237 -0.548706  0.58348
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## RMSE: 0.376291     Adj. R2: 0.81383
##                  Within R2: 1.582e-4

##
## Robustness check (excluding 2016-2018):

## OLS estimation, Dep. Var.: log_value_psf
## Observations: 9,269
## Fixed-effects: pin: 458,  tax_year: 24
## Standard-errors: Clustered (pin)
##                 Estimate Std. Error  t value Pr(>|t|)
## treatment:post -0.031545   0.052101 -0.60546  0.54517
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## RMSE: 0.380864     Adj. R2: 0.808532
##                  Within R2: 2.405e-4
```

4.2 Coefficient Comparison

```
##                    Specification    Coef     SE   Pval     N  CI_low CI_high
##                           <char>   <num>  <num>  <num> <int>   <num>   <num>
## 1:              Main (all years) -0.0259 0.0472 0.5835 10608 -0.1185  0.0667
## 2: Robustness (excl. 2016-2018) -0.0315 0.0521 0.5452  9269 -0.1337  0.0706
```

5 Event Study

The event study uses the robustness sample (2016-2018 excluded) with year −4 (2015) as the reference period. Pre-period coefficients near zero support the parallel trends assumption.

```
## OLS estimation, Dep. Var.: log_value_psf
## Observations: 9,269
## Fixed-effects: pin: 458,  tax_year: 24
## Standard-errors: Clustered (pin)
##                          Estimate Std. Error   t value  Pr(>|t|)
## year_rel::-20:treatment -0.025038   0.109475 -0.228705 0.8192002
## year_rel::-19:treatment  0.105855   0.093123  1.136727 0.2562480
## year_rel::-18:treatment  0.102692   0.096734  1.061595 0.2889802
## year_rel::-17:treatment  0.123869   0.073968  1.674634 0.0946903 .
## year_rel::-16:treatment  0.224960   0.063995  3.515285 0.0004831 ***
## year_rel::-15:treatment  0.080704   0.069383  1.163165 0.2453697
## year_rel::-14:treatment  0.093246   0.067560  1.380205 0.1681982
## year_rel::-13:treatment  0.067663   0.061407  1.101886 0.2710911
## year_rel::-12:treatment  0.054198   0.059983  0.903556 0.3667071
## year_rel::-11:treatment  0.062106   0.057199  1.085775 0.2781510
## year_rel::-10:treatment  0.098970   0.049988  1.979875 0.0483171 *
## year_rel::-9:treatment   0.023061   0.047796  0.482481 0.6296954
## year_rel::-8:treatment   0.030212   0.043578  0.693279 0.4884868
## year_rel::-7:treatment   0.050722   0.043699  1.160702 0.2463690
## year_rel::-6:treatment   0.079392   0.035497  2.236601 0.0257941 *
## year_rel::-5:treatment   0.101839   0.037075  2.746868 0.0062544 **
## year_rel::0:treatment    0.037246   0.046401  0.802700 0.4225651
## year_rel::1:treatment    0.030057   0.047849  0.628171 0.5302056
## year_rel::2:treatment    0.042415   0.060646  0.699391 0.4846638
## year_rel::3:treatment    0.112911   0.058710  1.923185 0.0550779 .
## year_rel::4:treatment    0.098262   0.059288  1.657358 0.0981333 .
## year_rel::5:treatment   -0.025530   0.060460 -0.422267 0.6730285
## year_rel::6:treatment   -0.026989   0.061654 -0.437743 0.6617791
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## RMSE: 0.380295     Adj. R2: 0.808624
##                  Within R2: 0.003224
```

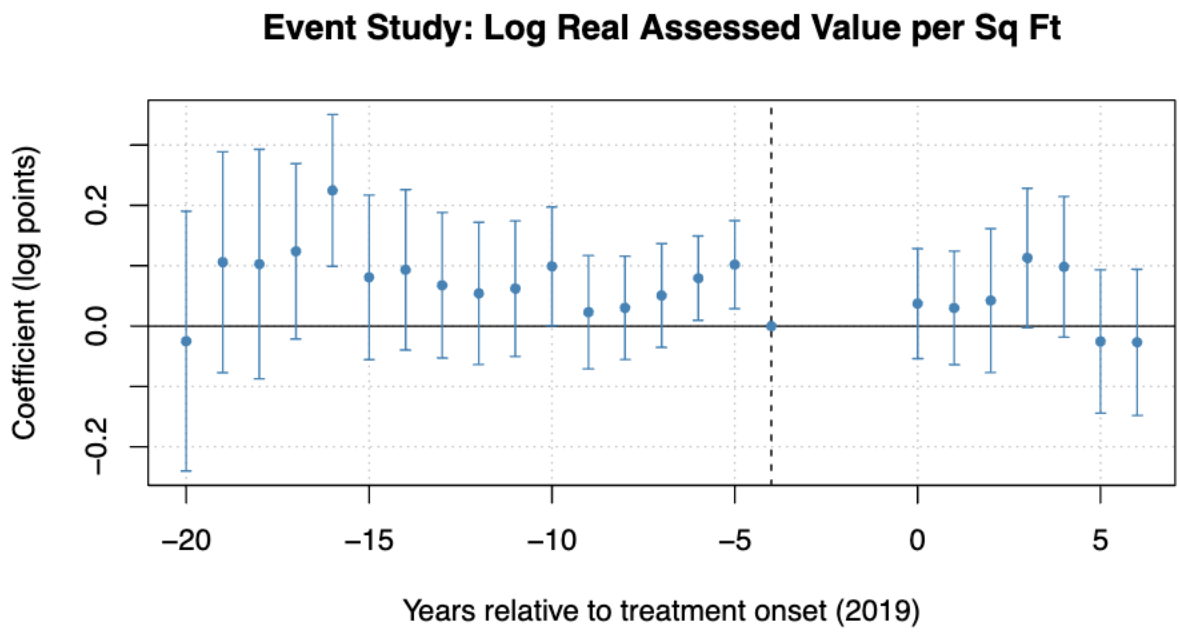

Figure 8: Event study coefficients for log real assessed value per sq ft. Reference year is 2015 (year -4). Bands show 95% confidence intervals.

6 Appendix: OLS Models

Included for reference only. The TWFE results in Section 4 are the preferred specification; OLS conflates within- and between-unit variation.

```
## Main specification:

##
## Call:
## lm(formula = value_psf ~ treatment * post, data = did_data_full)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -36.558 -16.590  -4.186  11.109  97.720
##
## Coefficients:
##                Estimate Std. Error t value Pr(>|t|)
## (Intercept)     27.8026     0.2865  97.027  < 2e-16 ***
## treatment        3.7852     0.6236   6.070 1.32e-09 ***
## post             7.5498     0.5286  14.284  < 2e-16 ***
## treatment:post  -0.8076     1.1453  -0.705    0.481
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## Residual standard error: 22.02 on 10604 degrees of freedom
## Multiple R-squared:  0.02703,    Adjusted R-squared:  0.02676
## F-statistic: 98.21 on 3 and 10604 DF,  p-value: < 2.2e-16
```

```
##
## Call:
## lm(formula = log_value_psf ~ treatment * post, data = did_data_full)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3.2544 -0.4927  0.1594  0.6036  1.8536
##
## Coefficients:
##                Estimate Std. Error t value Pr(>|t|)
## (Intercept)     2.97887    0.01144 260.327   <2e-16 ***
## treatment       0.22236    0.02490   8.929   <2e-16 ***
## post            0.28940    0.02111  13.711   <2e-16 ***
## treatment:post -0.03124    0.04574  -0.683    0.495
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.8792 on 10604 degrees of freedom
## Multiple R-squared:  0.03049,    Adjusted R-squared:  0.03021
## F-statistic: 111.2 on 3 and 10604 DF,  p-value: < 2.2e-16
```

```
##
## Robustness (excluding 2016-2018):
```

```
##
## Call:
## lm(formula = value_psf ~ treatment * post, data = did_data_clean)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -36.558 -16.442  -4.132  11.050  98.055
##
## Coefficients:
##                Estimate Std. Error t value Pr(>|t|)
## (Intercept)     27.2320     0.3159  86.200  < 2e-16 ***
## treatment        4.1968     0.6887   6.094 1.14e-09 ***
## post             8.1203     0.5448  14.904  < 2e-16 ***
## treatment:post  -1.2192     1.1816  -1.032    0.302
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 22 on 9265 degrees of freedom
```

```
## Multiple R-squared:  0.03258,    Adjusted R-squared:  0.03226
## F-statistic:   104 on 3 and 9265 DF,  p-value: < 2.2e-16
```

```
##
## Call:
## lm(formula = log_value_psf ~ treatment * post, data = did_data_clean)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3.2544 -0.4951  0.1584  0.5978  1.8756
##
## Coefficients:
##                Estimate Std. Error t value Pr(>|t|)
## (Intercept)     2.95500    0.01260 234.471   <2e-16 ***
## treatment       0.23596    0.02747   8.589   <2e-16 ***
## post            0.31328    0.02174  14.413   <2e-16 ***
## treatment:post -0.04484    0.04714  -0.951    0.341
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.8778 on 9265 degrees of freedom
## Multiple R-squared:  0.03619,    Adjusted R-squared:  0.03588
## F-statistic:   116 on 3 and 9265 DF,  p-value: < 2.2e-16
```